OpenClaw Temperature and Humidity Dashboard Tutorial

> Before starting, please complete the [01-Hardware Configuration], [02-openclaw API-KEY Configuration], [03-openclaw Model Switching], [04-AI Voice Interaction Configuration], [05-Three Dialogue Solution Configuration Tests] under the `Prerequisites` directory. If you are currently only using TUI / whatsapp, you can skip `04` for now. This article assumes the basic environment has been prepared and only expands on the temperature and humidity dashboard project itself.

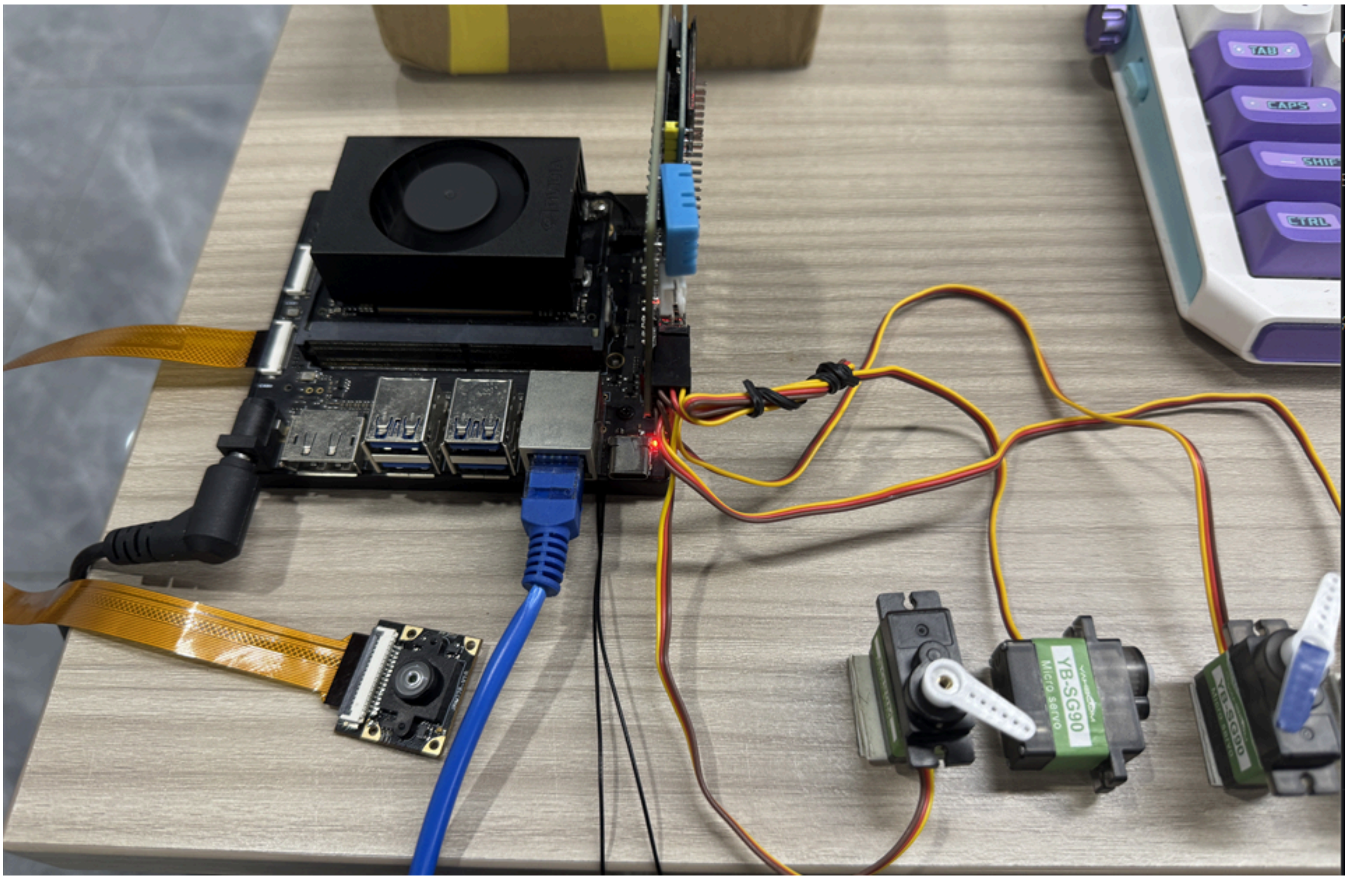
1. Tutorial Objectives

This article demonstrates OpenClaw's multi-peripheral coordination capability in desktop dashboard scenarios through the combination of "sensor + display + indication". After completing this tutorial, readers can:

- Understand how to combine DHT, OLED, RGB, and servo into a temperature and humidity dashboard
- Communicate with OpenClaw through three methods: whatsapp, TUI, and voice code
- Complete four project scenarios: real-time readings, OLED indicator panel, RGB status light, servo pointer mapping

2. Hardware Preparation

- OpenClaw mainboard
- DHT temperature and humidity sensor
- OLED display
- RGB light strip / RGB light bead module
- `[Optional] AI voice module + speaker`
- Wiring diagram



3. Software Preparation

Code entry points that will be used in the tutorial

- Temperature and humidity reading command: `/home/jetson/hardware_hal_demo/dht`
- OLED control command: `/home/jetson/hardware_hal_demo/oled`
- RGB control command: `/home/jetson/hardware_hal_demo/rgb`
- Servo control command: `/home/jetson/hardware_hal_demo/cli/servo`
- Voice entry: `/home/jetson/voice_to_openclaw/src/main.py`

This project is very close to the AI weather station, but the focus is different:

- AI weather station emphasizes more on "broadcast" and "explanation"
- Temperature and humidity dashboard emphasizes more on "display" and "status mapping"

4. Test Project Basic Capabilities

Before starting this project, please complete one `openclaw tui` basic dialogue self-check; if you haven't done it yet, see [05-Three Dialogue Solution Configuration Tests]. After passing, enter temperature and humidity dashboard specific testing.

It is recommended to first enter OpenClaw TUI:

```
openclaw tui
```

```
openclaw tui - ws://127.0.0.1:18789 - agent main - session main

session agent:main:main
connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens 0/1.0m (0%)
```

Then test the following types of commands:

- Read current temperature and humidity
- Display current temperature and humidity on OLED
- If environment is normal, light green
- Turn servo to 90 degrees

5. Three Solutions for Communicating with OpenClaw

Solution	Suitable Scenarios	Advantages	Recommended Positioning
whatsapp	Remote status viewing, classroom demonstrations	Convenient for remote reading of current panel data	Demonstrate "remote monitoring panel"
TUI	Local debugging, mapping verification	Easiest to troubleshoot reading and action relationships	First run through visualization chain
Voice code	Voice broadcast, desktop assistant	Closer to daily usage method	As complete demonstration solution

If optional AI voice module + speaker are selected, whatsapp dialogue, tui, whatsapp dialogue need to broadcast OpenClaw's reply content. You can run the following program in terminal. AI voice module dialogue does not need to run repeatedly

- Modify whether to enable broadcast parameter

```
/home/jetson/voice_to_openclaw/.env
```

```
ENABLE_TTS=true  #This parameter controls whether to broadcast. After modification and saving, run the following program in terminal
```

- Terminal input

```
cd ~/voice_to_openclaw
source .venv/bin/activate
python -m src.openclaw_session_tts
```

```
(.venv) jetson@yahboom:~/voice_to_openclaw$ python -m src.openclaw_session_tts
[系统] OpenClaw 会话播报器已启动
[系统] 监听目录=/home/jetson/.openclaw/agents/main/sessions
[系统] 监听渠道=feishu,tui,whatsapp
[播报] 开始播报 tui 回复
```

5.1 whatsapp Solution

For whatsapp connection and robot creation steps, please directly refer to [05-Three Dialogue Solution Configuration Tests]. After creation, you can remotely view dashboard status in whatsapp, for example:

- Temperature and humidity dashboard

20:36

< 2 +86 153 3885 7526 (你)
给自己发消息

warm or if temperature continues to rise

Would you like me to set up a reminder if the temperature exceeds 30°C?

Note: I did not schedule a reminder in this turn, so this will not trigger automatically.
20:34

Temperature and humidity dashboard
20:35

Temperature and humidity dashboard is now active!

📊 **Current Readings:**

- **Temperature:** 29.7°C (85.5°F)
- **Humidity:** 54%

📟 **OLED Display:** Showing temp and humidity values

💡 **RGB Indicator:** Red blinking light (speed 5)

- This indicates the temperature is **above 28°C** (warm/hot zone)

Status: The room is quite warm. Consider turning on a fan or AC if you're feeling uncomfortable!
20:35

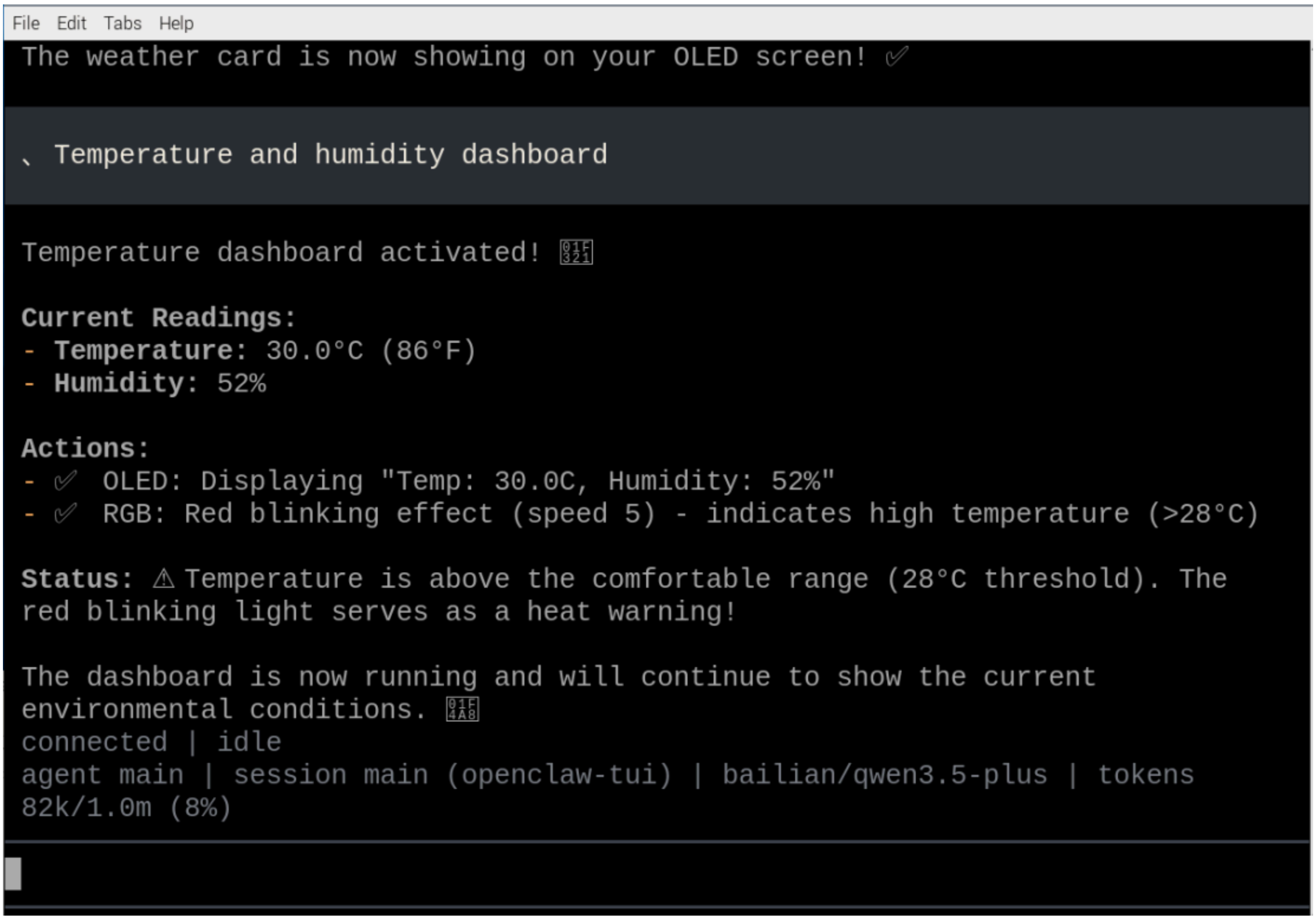
5.2 TUI Solution

If you have completed the TUI self-check in Chapter 0, directly enter terminal text dialogue:

```
openclaw tui
```

Then enter project-related commands, for example:

- Temperature and humidity dashboard

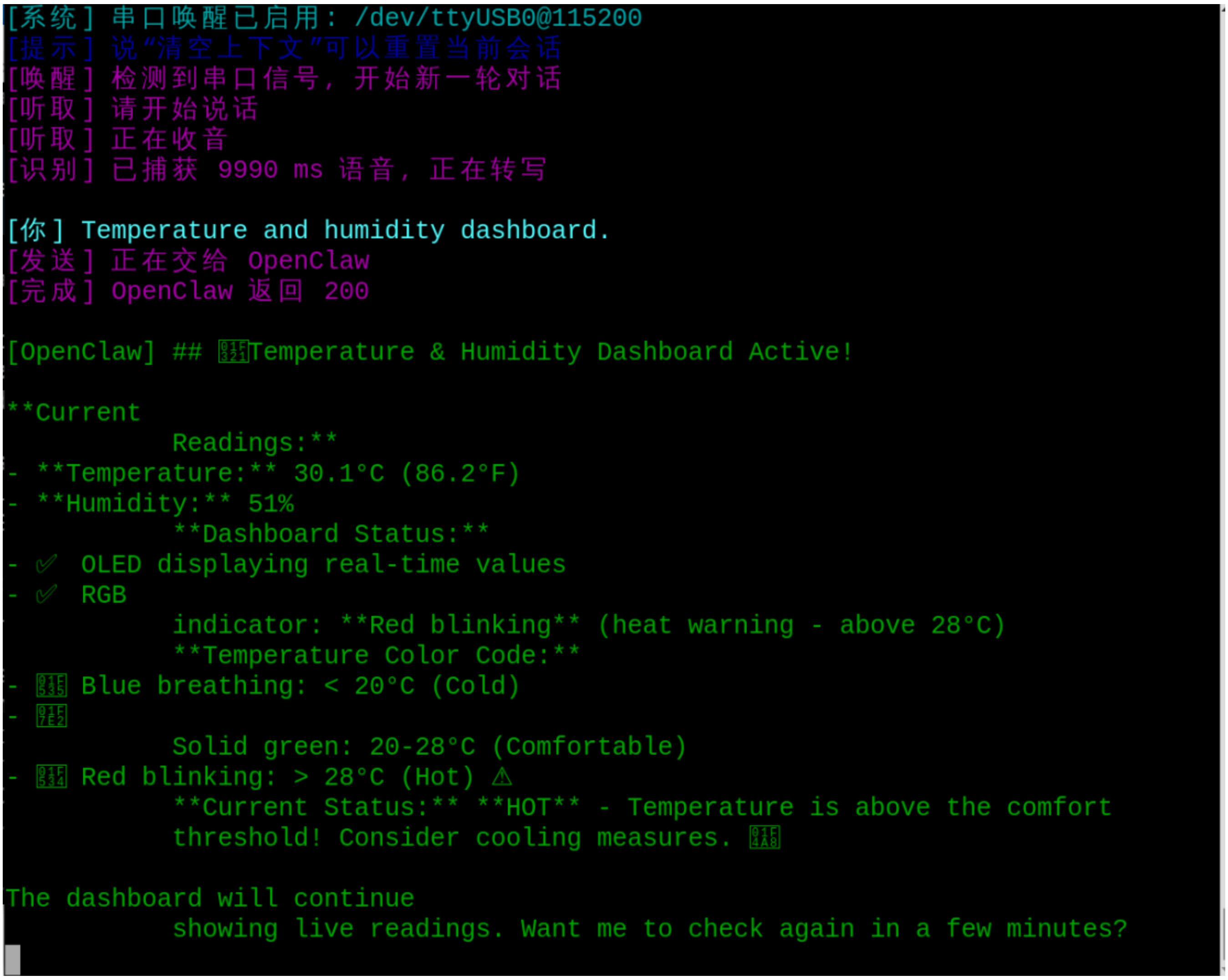


5.3 Voice Code Solution

```
Requires optional AI voice interaction module + speaker
```

For voice module installation, environment configuration, and basic wake-up process, please directly refer to [04-AI Voice Interaction Configuration]. After waking up, you can directly say control content, for example:

- Temperature and humidity dashboard



```
[系统] 串口唤醒已启用: /dev/ttyUSB0@115200
[提示] 说"清空上下文"可以重置当前会话
[唤醒] 检测到串口信号，开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 9990 ms 语音，正在转写

[你] Temperature and humidity dashboard.
[发送] 正在交给 OpenClaw
[完成] OpenClaw 返回 200

[OpenClaw] ## Temperature & Humidity Dashboard Active!

**Current
           Readings:**
- **Temperature:** 30.1°C (86.2°F)
- **Humidity:** 51%
           **Dashboard Status:**
- ✅ OLED displaying real-time values
- ✅ RGB
           indicator: **Red blinking** (heat warning - above 28°C)
           **Temperature Color Code:**
- Blue breathing: < 20°C (Cold)
-
           Solid green: 20-28°C (Comfortable)
- Red blinking: > 28°C (Hot) ⚠
           **Current Status:** **HOT** - Temperature is above the comfort
           threshold! Consider cooling measures.

The dashboard will continue
           showing live readings. Want me to check again in a few minutes?
```

6. Project Implementation

Note: The following are all dialogue demonstrations using `openclaw tui` method. You can also choose whatsapp or voice method to complete.

6.1 Scenario One: Real-time Readings

Experimental Objective

Read current temperature and humidity, completing the basic data input for the dashboard.

Effect Description

OpenClaw will first obtain temperature and humidity readings. These values are the basis for subsequent OLED, RGB, and servo mapping.

Example Instructions

- Read current temperature and humidity
- What is the current temperature and humidity
- Help me refresh the current readings

6.2 Scenario Two: OLED Indicator Panel

Experimental Objective

Display the current temperature and humidity results as concise OLED dashboard content.

Effect Description

Here it is suitable to use `metrics` or `dht_oled` method, combining `Temp` and `Hum` into a compact screen area, forming a small data panel.

Example Instructions

- Display Temp and Hum on OLED
- Make a simple temperature and humidity panel
- Write current temperature and humidity to the screen

6.3 Scenario Three: RGB Status Light

Experimental Objective

Use RGB colors to express whether the current environment is normal, too hot, too cold, or too dry.

Effect Description

This step can make the dashboard more intuitive. For example, when normal, light green; when too hot, light yellow; when abnormal, light red; when too cold, light blue. You can see the status at a glance.

Example Instructions

- If environment is normal, light green
- If temperature is high, light yellow
- If humidity is abnormal, light red

6.4 Scenario Four: Servo Pointer Mapping

Experimental Objective

Map temperature or humidity to servo angle, letting the servo point to a position like a dashboard pointer.

Effect Description

For example, map the temperature range to 0 to 180 degrees. The higher the current temperature, the more the servo pointer points to the right. This effect is very suitable for classroom demonstrations, because "numbers become physical actions" will be very intuitive.

Example Instructions

- Use servo to represent current temperature like a pointer
- Map current temperature to servo angle
- Let the pointer point to current status